

Boost Converter Problems

1.

(20 marks) The circuit shown in Figure 1 is a boost converter. It is operating with continuous inductor current, with duty cycle D and switching frequency $f_s=20\text{kHz}$. The input voltage is 5 volts, and the output voltage is 12 volts.

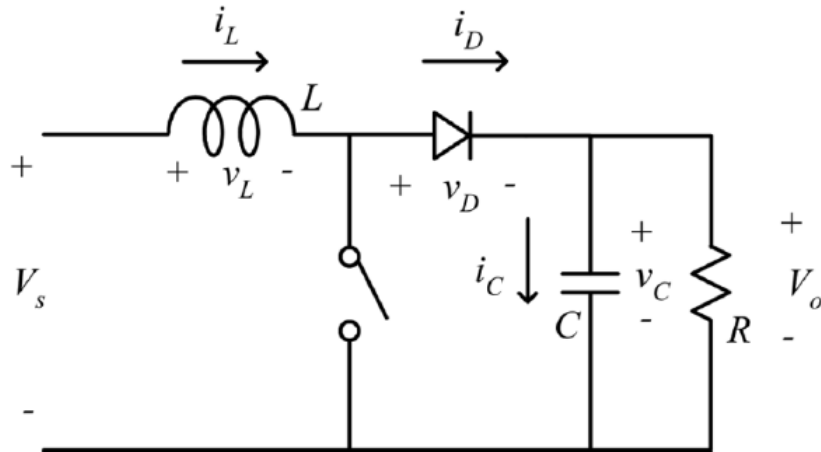


Figure 1: Boost converter

- Carefully draw waveforms showing inductor voltage, inductor current, diode current and capacitor ripple voltage. Make sure the timing is marked clearly.
- Showing your working, derive an expression for the step-up ratio of the converter as a function of the duty cycle D .
- Show that the value of load resistance that results in the inductor current becoming just discontinuous can be written as

$$R = \frac{2f_s L}{D(1-D)^2}.$$

- If the inductor is in continuous conduction by a reasonable margin, derive an expression for the output voltage ripple as a function of component values, duty cycle D , switching frequency f_s , and supply voltage V_s .